

Algorithm for file updates in Python

Project description

In this scenario I am a security analyst tasked with making sure that only allowed IP addresses maintain access to a restricted subnetwork. As such, I have created a python function that regularly checks the allowed list and updates it.

1. Open the file that contains the allow list

```
In [1]: # Assign 'import_file' to the name of the file
import_file = "allow_list.txt"

# Assign 'remove_list' to a list of IP addresses that are no longer allowed to access restricted information.
remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Display 'import_file'
print(import_file)

# Display 'remove_list'
print(remove_list)

allow_list.txt
['192.168.97.225', '192.168.158.170', '192.168.201.40', '192.168.58.57']
```

2. Read the file contents

```
In [3]: # Assign 'import_file' to the name of the file
import_file = "allow_list.txt"

# Assign 'remove_list' to a list of IP addresses that are no longer allowed to access restricted information.
remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build 'with' statement to read in the initial contents of the file
with open(import_file, "r") as file:
    # Use '.read()' to read the imported file and store it in a variable named 'ip_addresses'
    ip_addresses = file.read()

# Display 'ip_addresses'
print(ip_addresses)

ip_address 192.168.25.60 192.168.205.12 192.168.6.9 192.168.52.90 192.168.90.124 192.168.186.176 192.168.133.188 1
192.168.203.198 192.168.218.219 192.168.52.37 192.168.156.224 192.168.60.153 192.168.69.116
```

3. Convert the string into a list

```
In [4]: # Assign `import_file` to the name of the file
import_file = "allow_list.txt"

# Assign `remove_list` to a list of IP addresses that are no longer allowed to access restricted information.
remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build `with` statement to read in the initial contents of the file
with open(import_file, "r") as file:

    # Use `.read()` to read the imported file and store it in a variable named `ip_addresses`
    ip_addresses = file.read()

# Use `.split()` to convert `ip_addresses` from a string to a list
ip_addresses = ip_addresses.split()

# Display `ip_addresses`
print(ip_addresses)

['ip_address', '192.168.25.60', '192.168.205.12', '192.168.6.9', '192.168.52.90', '192.168.90.124', '192.168.186.176', '192.168.133.188', '192.168.203.198', '192.168.218.219', '192.168.52.37', '192.168.156.224', '192.168.60.153', '192.168.69.116']
```

The file is opened as string, and I converted into a list for easier iteration using the `.split()` method.

4. Iterate through the remove list

```
In [5]: # Assign `import_file` to the name of the file

import_file = "allow_list.txt"

# Assign `remove_list` to a list of IP addresses that are no longer allowed to access restricted information.

remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build `with` statement to read in the initial contents of the file

with open(import_file, "r") as file:

    # Use `.read()` to read the imported file and store it in a variable named `ip_addresses`

    ip_addresses = file.read()

# Use `.split()` to convert `ip_addresses` from a string to a list

ip_addresses = ip_addresses.split()

# Build iterative statement
# Name loop variable `element`
# Loop through `ip_addresses`

for element in ip_addresses:

    # Display `element` in every iteration

    print(element)
```

```
ip_address
192.168.25.60
192.168.205.12
192.168.6.9
192.168.52.90
192.168.90.124
192.168.186.176
192.168.133.188
192.168.203.198
192.168.218.219
192.168.52.37
192.168.156.224
192.168.60.153
192.168.69.116
```

5. Remove IP addresses that are on the remove list

```
In [6]: # Assign `import_file` to the name of the file

import_file = "allow_list.txt"

# Assign `remove_list` to a list of IP addresses that are no longer allowed to access restricted information.

remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build `with` statement to read in the initial contents of the file

with open(import_file, "r") as file:

    # Use `.read()` to read the imported file and store it in a variable named `ip_addresses`

    ip_addresses = file.read()

# Use `.split()` to convert `ip_addresses` from a string to a list

ip_addresses = ip_addresses.split()

# Build iterative statement
# Name loop variable `element`
# Loop through `ip_addresses`

for element in ip_addresses:

    # Build conditional statement
    # If current element is in `remove_list`,

    if element in remove_list:

        # then current element should be removed from `ip_addresses`

        ip_addresses.remove(element)

# Display `ip_addresses`

print(ip_addresses)

['ip_address', '192.168.25.60', '192.168.205.12', '192.168.6.9', '192.168.52.90', '192.168.90.124', '192.168.186.176', '192.168.133.188', '192.168.203.198', '192.168.218.219', '192.168.52.37', '192.168.156.224', '192.168.60.153', '192.168.69.116']
```

Identified IP addresses from the remove_list are removed using the .remove() method.

6. Update the file with the revised list of IP addresses

```
In [10]: # Define a function named 'update_file' that takes in two parameters: 'import_file' and 'remove_list'
# and combines the steps you've written in this lab leading up to this

def update_file(import_file, remove_list):

    # Build 'with' statement to read in the initial contents of the file
    with open(import_file, "r") as file:

        # Use '.read()' to read the imported file and store it in a variable named 'ip_addresses'
        ip_addresses = file.read()

        # Use '.split()' to convert 'ip_addresses' from a string to a list
        ip_addresses = ip_addresses.split()

        # Build iterative statement
        # Name loop variable 'element'
        # Loop through 'ip_addresses'

        for element in ip_addresses:

            # Build conditional statement
            # If current element is in 'remove_list',

            if element in remove_list:

                # then current element should be removed from 'ip_addresses'
                ip_addresses.remove(element)

        # Convert 'ip_addresses' back to a string so that it can be written into the text file
        ip_addresses = " ".join(ip_addresses)

        # Build 'with' statement to rewrite the original file
        with open(import_file, "w") as file:

            # Rewrite the file, replacing its contents with 'ip_addresses'
            file.write(ip_addresses)

    # Call 'update_file()' and pass in "allow_list.txt" and a list of IP addresses to be removed
    update_file("allow_list.txt", remove_list)

    # Build 'with' statement to read in the updated file
    with open("allow_list.txt", "r") as file:

        # Read in the updated file and store the contents in 'text'
        text = file.read()

    # Display the contents of 'text'
    print(text)

ip address 192.168.25.60 192.168.205.12 192.168.6.9 192.168.52.90 192.168.90.124 192.168.186.176 192.168.133.188 19
2.168.203.198 192.168.218.219 192.168.52.37 192.168.156.224 192.168.60.153 192.168.69.116
```

Summary

The list of allowed IP addresses was opened in python and an algorithm iterated through to confirm whether or not barred IP addresses were found. Barred addresses were indeed identified and removed, and the list was updated. A new function was built to automate subsequent checks and updates to the allowed list.